

Exercise Set 8

3 November 2025

1 Introducing quantifiers

- Is this formula satisfiable?

$$\forall x p(x) \iff \neg p(f(x))$$

- Is this formula satisfiable? Is it valid?

$$\forall x \exists y p(x, y)$$

- We define $\text{FV}(\phi)$ as the set of free variables of the FOL formul ϕ , i.e. the variables that appear in ϕ but are not quantified (e.g. $\forall x p(x) \wedge g(y)$, then y is a free variable). Can you give an inductive definition of ϕ ?

- Transform the following FOL formula into its *universal form*

$$\exists x p(x) \iff \forall y \exists z q(y, z)$$

- Transform the following FOL formula into its *universal form*

$$\exists x (p(x) \implies r(x) \vee \forall y p(y))$$

- Is the following set of clauses satisfiable? (a and b are constants, x and y are variables)

$$\{P(f(y, f(a, y))) \vee P(f(x, z)), \neg P(f(x, f(y, a))) \vee \neg P(f(f(a, b), x), y)\}$$

- You are given the set of functions $\mathcal{F} = \{a, f\}$ (a is a constant function with no parameters, and f takes one parameter) and the set of predicates $\mathcal{P} = \{p\}$. We consider the following FOL formula:

$$\forall x \phi \quad \text{where } \phi = p(x) \iff \neg p(f(x))$$

Propose 3 substitutions θ s.t. $\phi.\theta$ is a closed instance of ϕ . For each one of them, write the corresponding $\phi.\theta$.

Can you propose an interpretation such that $\forall x \phi$ is satisfiable?

- In this exercise, we fix the domain, the set of functions and predicates $(\mathcal{D}, \mathcal{F}, \mathcal{P})$. We say that a formula is $\mathcal{DFP}$ -valid if and only if for every interpretation the formula is satisfied.

Let C be a clause and Π a substitution on the variables of C . $\Pi(C)$ is the result of substituting every variable x in C by $\Pi(x)$.

Prove that if C is $\mathcal{DFP}$ -valid, then $\Pi(C)$ is $\mathcal{DFP}$ -valid

If Π is a renaming, namely a substitution which associates to every variable a fresh variable, then prove the reverse implication